Discrete Structures: The Foundations – Logic and Proofs: Part 4

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Outline

- Propositional Logic
- Applications of Propositional Logic
- Propositional Equivalences
- Predicates and Quantifiers
- Nested Quantifiers
- Rules of Inference
- Introduction to Proofs

Predicate

- We have learned "$x > 2$" is not a proposition
  - x is a variable and we can't directly tell the truth value
- How to describe this logic?
  - We need **Predicate Logic**
- "$x > 2$" or "x is greater than 2" has two parts:
  - 1. The variable x, is the subject of the statement
  - 2. The **predicate** – "is greater than 2" – refers to **a property** that the subject of the statement can have. We can use P to denote the predicate
  - The entire statement can be $P(x)$: this is also said to be the value of the **propositional function** P at x.
    - Once a value is assigned to x, the statement becomes a proposition and has a truth value.

Predicate Example

- **Example**: let $P(x)$ denote the statement "$x > 3$." The truth values of $P(4)$ and $P(2)$ are:

Predicate Logic	Proposition	Truth Value
$P(4)$	$\mathbf{4} > 3$	True
$P(2)$	$\mathbf{2} > 3$	False

- **Example**: Let $A(x)$ denote the statement "Computer x is under attack by an intruder." Suppose that of the computers on campus, only CS2 and MATH1 are currently under attack by intruders. What are the truth values of *A*(*CS1*), *A*(*CS2*), and *A*(*MATH1*) ?

Predicate Logic	Proposition	Truth Value
A($\mathbf{CS1}$)	Computer *CS1* is under attack by an intruder.	False
A($\mathbf{CS2}$)	Computer *CS2* is under attack by an intruder.	True
A($\mathbf{MATH1}$)	Computer *MATH1* is under attack by an intruder.	True

N-place predicate

- In general, a statement involving the n variables $x_1, x_2, \cdots, x_n$ can be denoted by $P(x_1, x_2, \cdots, x_n)$
- A statement of the form $P(x_1, x_2, \cdots, x_n)$ is the value of the **propositional function** P at the n-tuple $(x_1, x_2, \cdots, x_n)$, and P is also called an n-**place predicate** or a n-**ary predicate**.
  - **Example**: let $Q(x, y)$ denote the statement "$x = y + 3$." What are the truth values of the propositions $Q(1,2)$ and $Q(3,0)$?

Predicate Logic	Proposition	Truth Value
$\mathbf{Q(1,2)}$	$1 = 2 + 3$	False
$\mathbf{Q(3,0)}$	$3 = 0 + 3$	True

Quantifiers

- We may need to think about the range of values, to which the predicate can be applied
  - This range is called the **domain**
- We may also think about the values that make the predicate/property true
  - We need quantification, e.g. *all*, *some*, *many*, *none*, and *few*
  - Two most important ones:
    - *all* : universal quantifier
    - *some* : existential quantifier

Universal Quantifier

- $\forall x P(x)$ can be used to denote "$P(x)$ is true for all values of x in the domain."
  - $\forall$ is called the ***universal quantifier***
    - **Example**: let $P(x)$ be the statement "$x + 1 > x$", then $\forall x P(x)$ is true, in the domain of all real numbers.
  - $\forall x P(x)$ reads as "for all x $P(x)$" or "for every x $P(x)$."
    - Other ways: "all of", "for each", "given any", "for arbitrary", "for any", etc.
  - A value for which $P(x)$ is false is called a **counter example** of $\forall x P(x)$.
    - One counterexample is enough to say $\forall x P(x)$ is false
    - **Example**: let $Q(x)$ be the statement "$x > 2$" because $Q(1)$ is false, $\forall x Q(x)$ is false, in the domain of all real numbers.

Existential Quantifier

- $\exists x P(x)$ can be used to denote "There exists an element x in the domain such that $P(x)$."
  - $\exists$ is called the ***existential quantifier***
    - **Example**: let $Q(x)$ be the statement "$x > 2$" because $Q(4)$ is true,$\exists x Q(x)$ is true, in the domain of all real numbers.
  - $\exists x P(x)$ reads as "there exists"
    - Other ways: "for some", "for at least one", "there is", etc.
  - $\exists x P(x)$ is false, when for any value of x, $P(x)$ is false.
    - **Example**: let $P(x)$ be the statement "$x + 1 = x$", then $\exists x P(x)$ is false, in the domain of all real numbers.

Translation from English

- **Example**: Express the statement "If a person is female and is a parent, then this person is someone's mother" as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.
  - $F(x)$ "x is female."
  - $P(x)$ "x is a parent."
  - $M(x, y)$ "x is the mother of y."
  - $\forall x((F(x) \wedge P(x)) \rightarrow \exists y M(x, y))$ or
    $\forall x \exists y((F(x) \wedge P(x)) \rightarrow M(x, y))$

Precedence of Quantifiers

- The quantifiers ∀ and ∃ have higher precedence than all other logical operators

Operator	Precedence	Explanation
∀, ∃	1	Highest Precedence
¬	2	
∧	3	
⊕	4	
∨	5	
→	6	
↔	7	Lowest Precedence

Negating Quantified Expressions 1

- **Example**: what's the negation of "Every student in your class has taken a course in calculus."
  - "It is not the case that every student in your class has taken a course in calculus."
  - "There is a student in your class who has not taken a course in calculus."
- **Example**: what's the negation of "There is an honest politician."
  - Let $H(x)$ denote "x is honest."
  - $\neg\exists x H(x)$: "It is not the case that there is an honest politician."
  - $\forall x \neg H(x)$: "Every politician is dishonest."
  - $\neg\exists x H(x) \equiv \forall x \neg H(x)$

Negating Quantified Expressions 2

- Based on the definitions:

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.

- Then we can have

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\neg\forall x P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.
$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x.

- This means

$$\neg\forall \boldsymbol{x} \boldsymbol{P}(\boldsymbol{x}) \equiv \exists \boldsymbol{x} \neg \boldsymbol{P}(\boldsymbol{x})$$

Negating Quantified Expressions 3

- We can do similar analysis

Statement	When True?	When False?
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.
$\neg\exists x P(x)$	$P(x)$ is false for every x.	There is an x for which $P(x)$ is true.
$\forall x P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\forall x \neg P(x)$	$P(x)$ is false for every x.	There is an x for which $P(x)$ is true.

- This means

$$\neg\exists \boldsymbol{x} \boldsymbol{P}(\boldsymbol{x}) \equiv \forall \boldsymbol{x} \neg \boldsymbol{P}(\boldsymbol{x})$$

- **De Morgan's Laws for Quantifiers**

Negation	Equivalent Statement	When Is Negation True?	When False?
$\neg\exists x P(x)$	$\forall x \neg P(x)$	For every x, $P(x)$ is false	There is an x, for which $P(x)$ is true.
$\neg\forall x P(x)$	$\exists x \neg P(x)$	There is an x, for which $P(x)$ is false.	$P(x)$ is true for every x.

Nested Quantifiers 1

- Examples
  - For any integers x and y, x + y = y + x
    - $\forall x \forall y P(x, y)$
  - Some politicians can fool some of the people.
    - $\exists x \exists y P(x, y)$
  - Everyone has a mother.
    - $\forall x \exists y P(x, y)$
  - There is someone loved by everyone.
    - $\exists x \forall y P(x, y)$
- Can the order of nested quantifiers switched?
  - Yes for $\forall x \forall y P(x, y)$ and $\exists x \exists y P(x, y)$
    - Still the same combination of x and y
  - No for $\forall x \exists y P(x, y)$ and $\exists x \forall y P(x, y)$
    - There is a mother for everyone.
    - Everyone loves someone.

Nested Quantifiers 2

Statement	When True?	When False?
$\forall x \forall y P(x,y)$ $\forall y \forall x P(x,y)$	$P(x,y)$ is true for every pair x,y.	There is a pair x,y for which $P(x,y)$ is false.
$\forall x \exists y P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	There is an x such that $P(x,y)$ is false for every y.
$\exists x \forall y P(x,y)$	There is an x for which $P(x,y)$ is true for every y.	For every x there is a y for which $P(x,y)$ is false.
$\exists x \exists y P(x,y)$ $\exists y \exists x P(x,y)$	There is a pair x,y for which $P(x,y)$ is true.	$P(x,y)$ is false for every pair x,y.

- The statements $\forall x \forall y P(x,y)$ and $\forall y \forall x P(x,y)$ are logically equivalent.
- The statements $\exists y \exists x P(x,y)$ and $\exists x \exists y P(x,y)$ are logically equivalent.
- The statements $\exists y \forall x P(x,y)$ and $\forall x \exists y P(x,y)$ are **not** logically equivalent.

Nested Quantifiers 3

- **Example**: $P(x, y)$ means "$x + y = 0$", where x and y are integers. Determine the truth value of:
  - $\exists x \forall y P(x, y)$: **false**
    - There is one value of x (call it x_0) that can be associated with any value of y to make $P(x, y)$ true
    - $x_0 + y = 0$, then how about we try $y = -x_0 + 1$? By substitution we will have $x_0 - x_0 + 1 = 0$, which can be further simplified as $1 = 0$. Clearly this false.
  - $\forall y \exists x P(x, y)$: **true**
    - For any value of y, we can find a value of x, to pair with it and make $P(x, y)$ true
    - For $y = y_1$, we can have $x = -y_1$ to make $P(-y_1, y_1)$ true
    - For $y = y_2$, we can have $x = -y_2$ to make $P(-y_2, y_2)$ true

Negating Nested Quantifiers

- Move negation inside quantifiers: $\neg\exists x\forall y P(x,y)$

Statement	Justification
$\neg\exists x\forall y P(x,y)$	Original input
$\equiv \neg\exists x Q(x,y)$	Substitution
$\equiv \forall x \neg Q(x,y)$	De Morgan's Laws for quantifiers
$\equiv \forall x \neg\forall y P(x,y)$	Substitution
$\equiv \forall x\exists y \neg P(x,y)$	De Morgan's Laws for quantifiers

  - Negation is used to flip the boolean/truth values
  - We need to flip the predicate: from $P(x,y)$ to $\neg P(x,y)$
  - We need to flip the quantifiers: from ∃ to ∀, and from ∀ to ∃
- Try other nested cases
  - E.g. $\neg\forall x\forall y P(x,y)$, $\neg\forall x\exists y P(x,y)$, $\neg\exists x\exists y P(x,y)$
  - $\exists x\exists y\neg P(x,y)$, $\exists x\forall y\neg P(x,y)$, $\forall x\forall y\neg P(x,y)$

Practices 1

- 1. Let $P(x)$ be the statement "$x = x^2$". If the domain consists of all integers, what are the truth values?

  - $P(0)$
  - $P(1)$
  - $P(2)$
  - $P(-1)$
  - $\exists x\ P(x)$
  - $\forall x P(x)$

Practices 2

- 1. Let $P(x)$ be the statement "$x = x^2$". If the domain consists of all integers, what are the truth values?

  - $P(0)$: true -- $0 = 0^2$
  - $P(1)$: true -- $1 = 1^2$
  - $P(2)$: false -- $2 \neq 2^2$
  - $P(-1)$: false -- $-1 \neq (-1)^2$
  - $\exists x\ P(x)$: true -- work for input 0
  - $\forall x P(x)$: false -- doesn't work for input -1

Practices 3

- 2. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives.
  - No one is perfect.
  - Not everyone is perfect.
  - At least one of your friends is perfect.
  - Everyone is your friend and is perfect.
  - Not everybody is your friend or someone is not perfect.

Practices 4

- 2. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives.
  - Let $P(x)$ be "x is perfect" and $F(x)$ be "x is your friend".
  - No one is perfect.
    - $\neg\exists x P(x)$ or $\forall x\ \neg P(x)$
  - Not everyone is perfect.
    - $\neg\forall x P(x)$ or $\exists x\ \neg P(x)$
  - At least one of your friends is perfect.
    - $\exists x\ (F(x) \wedge P(x))$
  - Everyone is your friend and is perfect.
    - $\forall x\ (F(x) \wedge P(x))$
  - Not everybody is your friend or someone is not perfect.
    - $(\neg\forall x\ F(x)) \vee (\exists x\ \neg P(x))$

Practices 5

- 3. Translate “there exist no integers a and b for which $18a + 6b = 1$”

Practices 6

- 3. Translate "there exist no integers a and b for which $18a + 6b = 1$"
  - "there doesn't exist integers a and b for which $18a + 6b = 1$"
  - "it's not the case that there exist integers a and b for which $18a + 6b = 1$" -- key: find the negation
  - $\neg\exists a\exists b(18a + 6b = 1) \equiv \forall a\forall b(18a + 6b \neq 1)$
    - -- key: De Morgan's Law

Practices 7

- 4. Let $L(x,y)$ be the statement "x loves y", where the domain for both x and y consists of all people in the world. Use quantifiers to express each statement.
  - Everybody loves Jerry.
  - Nobody loves everybody.
  - There is somebody whom no one loves.
  - Everyone loves himself or herself.

Practices 8

- 4. Let $L(x,y)$ be the statement "x loves y", where the domain for both x and y consists of all people in the world. Use quantifiers to express each statement.
  - Everybody loves Jerry.
    - $\forall x L(x,\text{Jerry})$
  - Nobody loves everybody.
    - $\neg\exists x \forall y L(x,y)$ or $\forall x \exists y \neg L(x,y)$
  - There is somebody whom no one loves.
    - $\exists x \forall y \neg L(y, x)$ or $\neg\forall x \exists y L(y, x)$
  - Everyone loves himself or herself.
    - $\forall x L(x, x)$

Extension 1

- 1. About domain
  - When the domain is changed, the quantification of $P(x)$ may change. So we must always specify the domain.
  - **Empty domain**
    - $\forall x P(x)$ is true: no counterexample.
    - $\exists x P(x)$ is false: no evidence
  - **Finite domain**
    - **Example**: let $P(x)$ be the statement "$x^2 < 10$" for the positive integers not exceeding 4, check $\forall x P(x)$
      - $\forall x P(x)$ is equal to $P(1) \wedge P(2) \wedge P(3) \wedge P(4)$
      - $P(4)$ is a counterexample, so $\forall x P(x)$ is false
    - **Example**: let $P(x)$ be the statement "$x^2 > 10$" for the positive integers not exceeding 4, check $\exists x P(x)$
      - $\exists x P(x)$ is equal to $P(1) \vee P(2) \vee P(3) \vee P(4)$
      - $P(4)$ is true, so $\exists x P(x)$ is true

Extension 2

- 2. Abbreviated notation for restricted domain
  - Assume we are talking about real numbers
    - A. "The square of any negative real number is positive."
      - Abbreviated notation: $\forall x < 0(x^2 > 0)$
        - Much easier to work with restricted domain
      - Complete version: $\forall x(x < 0 \rightarrow x^2 > 0)$
        - Why implication in the middle?
          - Think about the truth value: if x >= 0, we can't deny any result, so we can't say the truth value should be false
        - "If a real number is negative, then its square is positive."
    - B. "The cube of every nonzero real number is nonzero. "
      - $\forall y \neq 0(y^3 \neq 0)$ or $\forall y(y \neq 0 \rightarrow y^3 \neq 0)$
    - C. "There is a positive square root of 2."
      - $\exists z > 0(z^2 = 2)$ or $\exists z(z > 0 \wedge z^2 = 2)$
        - "There is a real number that is positive and is a square root of 2."

Extension 3

- 3. Comprehensive problems
  - A. Do you think $\exists x \forall y P(x,y) \rightarrow \forall y \exists x P(x,y)$ is always true / a tautology?
    - Yes. Check by definitions.
    - $\forall y \exists x P(x,y) \rightarrow \exists x \forall y P(x,y)$ will be false (otherwise, LHS≡RHS which is wrong, because we can't randomly switch the order of two different nested quantifiers)
  - B. Show that

$$\neg \forall x \big(P(x) \rightarrow Q(x)\big) \equiv \exists x \big(P(x) \wedge \neg Q(x)\big)$$

Expression	Justification
$\neg \forall x (P(x) \rightarrow Q(x))$	Initial assumption
$\equiv \exists x (\neg (P(x) \rightarrow Q(x)))$	De Morgan's Law
$\equiv \exists x (P(x) \wedge \neg Q(x))$	Tautology + De Morgan's Law

Extension 4

- 4. Uniqueness Quantifiers
  - There is exactly one person whom everybody can fool
    - Two objects x, y
    - Property: $F(x, y)$ for "x can fool y"
    - $\exists \boldsymbol{y}(\forall \boldsymbol{x} \boldsymbol{F}(\boldsymbol{x}, \boldsymbol{y}) \wedge \forall \boldsymbol{z}(\forall \boldsymbol{x} \boldsymbol{F}(\boldsymbol{x}, \boldsymbol{z}) \rightarrow \boldsymbol{z} = \boldsymbol{y}))$
  - We can use uniqueness quantifiers to make it easier
    - $\exists! x P(x)$ or $\exists_1 x P(x)$
      - This means "There exists a unique x such that $P(x)$ is true."
    - So for the example on the top, we can have: $\exists_1 \boldsymbol{y}\ \forall x\ \boldsymbol{F}(\boldsymbol{x}, \boldsymbol{y})$
  - Similarly, we can have "exactly two" as $\exists_2 x P(x)$